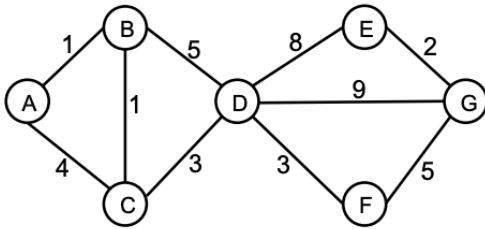


Practice Problems:

Question 1



Node	h_1	h_2
A	9.5	10
B	9	12
C	8	10
D	7	8
E	1.5	1
F	4	4.5
G	0	0

Consider the state space graph shown above. A is the start state and G is the goal state. The costs for each edge are shown on the graph. Each edge can be traversed in both directions. Note that the heuristic h_1 is consistent but the heuristic h_2 is not consistent.

(a) Possible paths returned

For each of the following graph search strategies (*do not answer for tree search*), mark which, if any, of the listed paths it could return. Note that for some search strategies the specific path returned might depend on tie-breaking behavior. In any such cases, make sure to mark *all* paths that could be returned under some tie-breaking scheme.

Search Algorithm	A-B-D-G	A-C-D-G	A-B-C-D-F-G
Depth first search	x	x	x
Breadth first search	x	x	
Uniform cost search			x
A* search with heuristic h_1			x
A* search with heuristic h_2			x

The return paths depend on tie-breaking behaviors so any possible path has to be marked. DFS can return any path. BFS will return all the shallowest paths, i.e. A-B-D-G and A-C-D-G. A-B-C-D-F-G is the optimal path for this problem, so that UCS and A* using consistent heuristic h_1 will return that path. Although, h_2 is not consistent, it will also return this path.

(b) Heuristic function properties

Suppose you are completing the new heuristic function h_3 shown below. All the values are fixed except $h_3(B)$.

Node	A	B	C	D	E	F	G
h_3	10	?	9	7	1.5	4.5	0

For each of the following conditions, write the set of values that are possible for $h_3(B)$. For example, to denote all non-negative numbers, write $[0, \infty]$, to denote the empty set, write $\emptyset$, and so on.

(i) What values of $h_3(B)$ make h_3 admissible

To make h_3 admissible, $h_3(B)$ has to be less than or equal to the actual optimal cost from B to goal G, which is the cost of path B-C-D-F-G, i.e. 12. The answer is $0 \leq h_3(B) \leq 12$

(ii) What values of $h_3(B)$ make h_3 consistent?

All the other nodes except node B satisfy the consistency conditions. The consistency conditions that do involve the state B are:

$$h(A) \leq c(A, B) + h(B) \quad h(B) \leq c(B, A) + h(A)$$

$$h(C) \leq c(C, B) + h(B) \quad h(B) \leq c(B, C) + h(C)$$

$$h(D) \leq c(D, B) + h(B) \quad h(B) \leq c(B, D) + h(D)$$

Filling in the numbers shows this results in the condition: $9 \leq h_3(B) \leq 10$

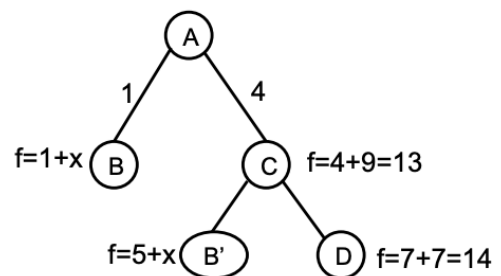
(iii) [2 pts] What values of $h_3(B)$ will cause A* graph search to expand node A, then node C, then node B, then node D in order?

The A* search tree using heuristic h_3 is on the right. In order to make A* graph search expand node A, then node C, then node B, suppose $h_3(B) = x$, we need

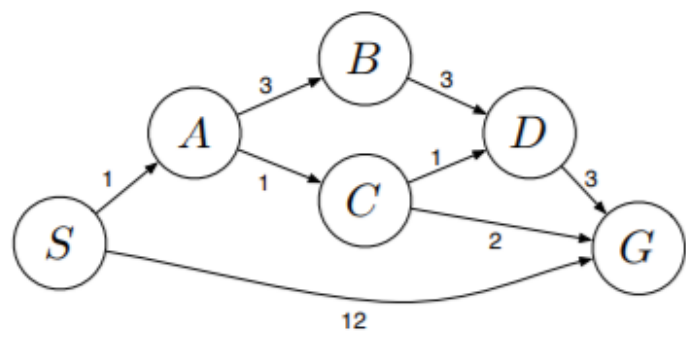
$$1 + x > 13$$

$$5 + x < 14 \quad (\text{expand } B') \quad \text{or} \quad 1 + x < 14 \quad (\text{expand } B)$$

so we can get $12 < h_3(B) < 13$



Question 2



Answer the following questions about the search problem shown above. Break any ties alphabetically. For the questions that ask for a path, please give your answers in the form ‘S – A – D – G.’

(a) (2 pt) What path would breadth-first graph search return for this search problem?

S – G

(b) (2 pt) What path would uniform cost graph search return for this search problem?

S – A – C – G

(c) (2 pt) What path would depth-first graph search return for this search problem?

~~S – A – B – D – G~~ S G

(d) (2 pt) What path would A* graph search, using a consistent heuristic, return for this search problem?

S – A – C – G

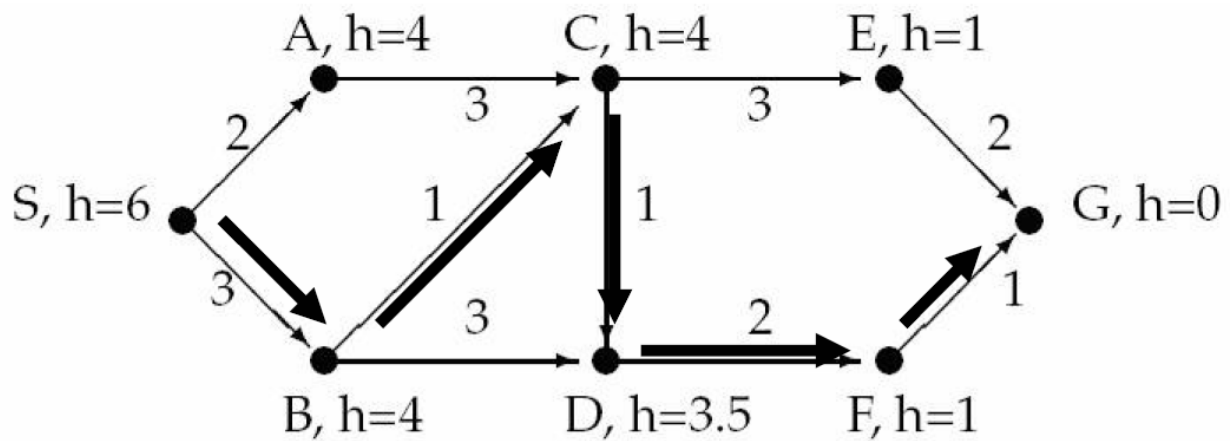
(e) (4 pt) Consider the heuristics for this problem shown in the table below.

State	h_1	h_2
S	5	4
A	3	2
B	6	6
C	2	1
D	3	3
G	0	0

- i. (1 pt) Is h_1 admissible? Yes No
- ii. (1 pt) Is h_1 consistent? Yes No
- iii. (1 pt) Is h_2 admissible? Yes No
- iv. (1 pt) Is h_2 consistent? Yes No

Question 3

Suppose we want to use the A* algorithm on the graph below to find the shortest path from node S to node G. Each node is labeled by a capital letter and the value of a heuristic function. Each edge is labeled by the cost to traverse that edge.



For this problem:

- Perform the A* algorithm on this graph, filling in the table below. You should not need all the lines in the table. Indicate the f, g, and h values of each node on the queue as shown in the first row of the table.
- Show the path found by the A* algorithm on the graph above.

iteration	node expanded	Priority queue at end of this iteration
0		$S = 0 + 6 = 6$ (i.e. $S = g(S) + h(S) = f(S)$)
1	S	$A = 2 + 4 = 6$; $B = 3 + 4 = 7$
2	A	$B = 7$, $C = 2 + 3 + 4 = 9$
3	B	$C = 3 + 1 + 4 = 8$, $D = 3 + 3 + 3.5 = 9.5$
4	C	$E = 4 + 3 + 1 = 8$, $D = 4 + 1 + 3.5 = 8.5$
5	E	$D = 8.5$, $G = 7 + 2 + 0 = 9$
6	D	$F = 5 + 2 + 1 = 8$, $G = 9$
7	F	$G = 7 + 1 + 0 = 8$
8	G	
9		
10		

Question 4

Consider the following example. Let A be the starting node and K be its destination. Find the most optimal path from starting state to goal state by using A* searching algorithm. Heuristic table is shown below.

Nodes	Cost
A	11
B	3
C	1
D	4
E	5
F	8
G	9
H	3
I	7
J	2
K	6
M	2
N	3
O	3
P	1

